Sophie Huiberts

Curriculum Vitae, January 2026

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Work experience

Oct 2023 – present **Chargée de Recherche**, *CNRS, LIMOS laboratory, Clermont-Ferrand*

Jul 2022 – Aug 2023 **Postdoctoral fellow**, *Computer Science Department, Columbia University*

Dec 2017 – Jun 2022 **PhD candidate**, *Centrum Wiskunde & Informatica*
- Supervisor: Daniel Dadush. Promotor: Gunther Cornelissen
- Defense date: May 16th 2022
- Thesis: Geometric Aspects of Linear Programming: Shadow Paths, Central Paths, and a Cutting Plane Method (Stieltjes Prize, Gijs de Leve Prize)

Feb 2017 – Nov 2017 **Research trainee**, *Centrum Wiskunde & Informatica*
Supervisor: Daniel Dadush. Subject: smoothed analysis of the simplex method.

Sep 2016 – Aug 2017 **Student board member**, *Department of Mathematics, Utrecht University*
Representing students' interests on education and policy matters. 0.4 fte.

Education

Feb 2016 – Jan 2018 **Master Mathematical Sciences**, *Utrecht University*

Sep 2012 – Aug 2017 **Bachelor Computer Science**, *Utrecht University*

Sep 2012 – Feb 2016 **Bachelor Mathematics**, *Utrecht University*

Honours and Awards

2024 – 2028 **ANR JCJC Grant**, *Agence Nationale de la Recherche*, Project: Towards Testable Theories of Linear Programming

Jan 2024 **Gijs de Leve Prize**, *Landelijk Netwerk Mathematische Besliskunde*, best Dutch PhD thesis in mathematics of operations research, 2020–2023

Apr 2023 **Stieltjes Prize**, *Platform Wiskunde Nederland*, best Dutch PhD thesis in mathematics, academic year 2021-2022

2022 – 2023 **Junior Fellowship**, *Simons Society of Fellows*

Jun 2021 **Rising Star Speaker**, *TCS Women Spotlight Workshop*, ACM Symposium on Theory of Computing (STOC)

Sep 2019 **Attendee**, *Heidelberg Laureate Forum*, selective conference admitting 100 young researchers across mathematics and computer science

Featured in

- *Researchers Discover the Optimal Way To Optimize*, Steve Nadis, Quanta Magazine, October 2026. Syndicated in Wired, December 2026.
- *Women in Optimization: Dr. Sophie Huiberts*, Elisabeth Rodriguez Heck, Gurobi Optimization, July 2024.
- *Proof by example: Sophie Huiberts*, Clara Stegehuis and Francesca Arici, Nieuw Archief voor Wiskunde, December 2022.
- *Mathématiques: lumière sur les étranges performances de l'algorithme du simplexe*, Clémentine Laurens, Le Monde, 4 October 2022.
- *Het diversiteitsprobleem is geen vrouwenprobleem*, Ans Hekkenberg, CWI 75th anniversary special of New Scientist, 2021.

Publications

In my field of research, authors are listed in alphabetical order.

[1] E. Bach, Y. Disser, S. Huiberts, and N. Mosis, "An unconditional lower bound for the active-set method in convex quadratic maximization," in *Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, Society for Industrial and Applied Mathematics, Jan. 2026, pp. 314–327. DOI: 10.1137/1.9781611978971.14. arXiv: 2507.16648.

[2] E. Bach, A. E. Black, S. Huiberts, and S. Kafer, *Beyond smoothed analysis: Analyzing the simplex method by the book*, 2025. arXiv: 2510.21613.

[3] E. Bach and S. Huiberts, "Optimal smoothed analysis of the simplex method," in *66th IEEE Symposium on Foundations of Computer Science (FOCS)*, 2025. arXiv: 2504.04197.

[4] S. Huiberts, Y. T. Lee, and X. Zhang, "Upper and lower bounds on the smoothed complexity of the simplex method," *TheoretiCS*, vol. 4, 23, Oct. 2025, preliminary version in STOC 2023, ISSN: 2751-4838. DOI: 10.46298/theoretics.25.23. arXiv: 2211.11860.

[5] S. Borst, D. Dadush, S. Huiberts, and D. Kashaev, "A nearly optimal randomized algorithm for explorable heap selection," *Mathematical Programming*, vol. 210, no. 1–2, pp. 75–96, Nov. 2024, preliminary version in IPCO 2023, ISSN: 1436-4646. DOI: 10.1007/s10107-024-02145-5. arXiv: 2210.05982.

[6] D. Dadush, C. Hojny, S. Huiberts, and S. Weltge, "A simple method for convex optimization in the oracle model," *Mathematical Programming*, Aug. 2023, preliminary version in IPCO 2022. DOI: 10.1007/s10107-023-02005-8. arXiv: 2011.08557.

[7] D. Dadush, S. Huiberts, B. Natura, and L. A. Végh, "A scaling-invariant algorithm for linear programming whose running time depends only on the constraint matrix," *Mathematical Programming*, Apr. 2023, preliminary version in ACM STOC 2020. DOI: 10.1007/s10107-023-01956-2. arXiv: 1912.06252.

[8] G. Bonnet, D. Dadush, U. Grupel, S. Huiberts, and G. Livshyts, "Asymptotic Bounds on the Combinatorial Diameter of Random Polytopes," in *38th International Symposium on Computational Geometry (SoCG)*, vol. 224, 2022, 18:1–18:15. DOI: 10.4230/LIPIcs.SoCG.2022.18. arXiv: 2112.13027.

[9] S. Borst, D. Dadush, S. Huiberts, and S. Tiwari, "On the integrality gap of binary integer programs with Gaussian data," *Mathematical Programming*, 2022, preliminary version in IPCO 2021. DOI: 10.1007/s10107-022-01828-1. arXiv: 2012.08346.

[10] D. Dadush and S. Huiberts, "Smoothed analysis of the simplex method," in *Beyond Worst Case Analysis*, T. Roughgarden, Ed., Cambridge University Press, 2021, ch. 14, pp. 309–333. DOI: 10.1017/9781108637435.019. [Online]. Available: https://sophie.huiberts.me/files/bwca-chapter-simplex.pdf.

[11] D. Dadush and S. Huiberts, "A friendly smoothed analysis of the simplex method," *SIAM Journal on Computing*, vol. 49, no. 5, STOC18-449–499, 2020, preliminary version in ACM STOC 2018. DOI: 10.1137/18M1197205. arXiv: 1711.05667.

Supervision

- Doctoral Researchers
  - Eleon Bach. 2022-Present. TU Munich.
- Postdoctoral Researchers
  - François Lamothe. 2026-Present. LIMOS.
- Students
  - Guilherme Knupp Muniz. 2025. LIMOS.

Service to the community

2027 **Safety and Inclusion Committee**, *International Symposium on Mathematical Programming (ISMP)*

2026 **Co-organizer**, *Dagstuhl Seminar: Analysis of Algorithms Beyond the Worst Case*

2026 **Program Committee**, *Scandinavian Symposium on Algorithm Theory (SWAT)*

2026 **Program Committee**, *International Symposium on Theoretical Aspects of Computer Science (STACS)*

2025 – present **Editor**, *European Association for Theoretical Computer Science (EATCS) on Youtube*

2025 **PC & Local Chair**, *Mixed Integer Programming Workshop Europe (MIP Europe)*

2025 **Program Committee**, *Caleidoscope Research School in Computational Complexity*

2025 **Program Committee**, *Mixed Integer Programming Workshop (MIP)*

2024 – present **Editorial Board**, *Discrete Optimization*

2024 **Program Committee**, *International Colloquium on Automata, Languages and Programming (ICALP)*

2024 **Program Committee**, *Integer Programming and Combinatorial Optimization (IPCO)*

2023 – present **Scientific Committee**, *Groupe de Travail Complexité et Algorithmes*

Spring 2022 **Research Assessment Committee**, *Applied Mathematics TU Delft 2015–2020*

May 2018 – Mar 2022 **Works Council**, *Centrum Wiskunde & Informatica*
Chair from March 2021 – March 2022.

Nov 2017 – Jun 2022 **Local Representative**, *European Women in Mathematics The Netherlands*

Aug 2020 – Dec 2020 **Seminar Organizer**, *Probability, Geometry, and Computation in High Dimensions Student Seminar*, Simons Institute for the Theory of Computing

Oct 2019 – Apr 2020 **Communications Team**, *European Girls Mathematical Olympiad 2020*

Software

Oct 2019 – Dec 2023 **Lead Author**, *LaTeX in Slack*
Mathematical remote collaboration software that served over 3500 simultaneous installs. Browser extension that allows LaTeX formulas on chat service Slack.

Miscellaneous writing

- *Het Dieetprobleem*, STAtOR, July 2025.
- *Prizes and Prejudice*, Bulletin of the European Association for Theoretical Computer Science, October 2022.
- *Wanneer is Software Snel?*, AG Connect, November 2021.

Presentations

Lecture Series

- *Open Problems about the Simplex Method*, IPCO Summer School, July 1–2, 2024.

Keynote Lectures

- *Open Problems about the Simplex Method*
  - CIAC, Rome, June 10–12, 2025.
  - Workshop on Polytopes, February 24–28th, 2025.
  - Combinatorics and Geometry in Villetaneuse, November 7–8, 2024.
- *Smoothed Analysis of the Simplex Method: Upper and Lower Bounds*
  - Workshop Complexité et Algorithmes de CoA, September 19–20, 2023.
  - Mixed Integer Programming Workshop, May 22–25, 2023.
  - Aussois Combinatorial Optimization Workshop, January 9–13, 2023.

Contributed & Seminar Presentations

- *Everything Wrong with Smoothed Analysis*, Aussois Combinatorial Optimization Workshop, January 5–9, 2025.
- *Linear Programming: Algorithms and Ethics*, ENS Rennes, December 3rd, 2025.
- *Beyond Smoothed Analysis: Analyzing the Simplex Method by the Book*, ETH Zurich, November 18th, 2025.
- *The Diet Problem*, G-SCOP, October 9th, 2025.
- *Optimal Smoothed Analysis of the Simplex Method*
  - FOCS, YouTube, December 2025.
  - OPTIMA, Online, August 20th, 2025.
- *The Life and Work of Paula Harris*
  - Mixed Integer Programming, YouTube, August 2025.
  - The Simplex Method: Theory, Complexity, and Applications, June 26–27, 2025.
- *Open Problems about the Simplex Method*
  - UCLouvain, April 15th, 2025.
  - ETH Zurich, April 10th, 2025.
  - Séminaire du GT Graphes, March 20th, 2025.
  - LIRMM Montpellier, January 14th, 2025.
  - OMP, Antwerp, October 16th, 2024.
  - EPFL Bernoulli Center, August 19th, 2024.
  - The University of Edinburgh, May 22nd, 2024.
  - Séminaire Parisien d'Optimisation, May 6th, 2024.
  - Bocconi University, April 10th, 2024.
  - Journées SMAI-MODE, March 27–29, 2024.
  - Journées Nationales du GdR IFM, March 21st, 2024.
  - IRIF, February 27th, 2024.
  - Université Claude Bernard Lyon 1, February 12th, 2024.
  - Zuse Institut Berlin, February 7th, 2024.
  - FU Berlin, February 1st, 2024.
  - 27th Seminar of the POC group, December 15th, 2023.
  - CWI Networks & Optimization Seminar, November 15th, 2023.
- *A Nearly Optimal Randomized Algorithm for Explorable Heap Selection*
  - University of Washington, May 30th, 2023.
  - École Polytechnique, March 14th, 2023.
- *Smoothed Analysis of the Simplex Method: Upper and Lower Bounds*
  - SIAM Conference on Applied Algebraic Geometry, July 10–14, 2023.
  - Journées Polyèdres et Optimisation Combinatoire, June 28–30, 2023.
  - STOC, YouTube, June 2025.
  - Copenhagen-Jerusalem Combinatorics Seminar, May 4, 2023.
  - University of Massachusetts Amerherst Theory Seminar, April 4th, 2023.
  - ICERM Combinatorics and Optimization Workshop, March 27–31, 2023.

- CNRS LIMOS lab, March 9th, 2023.
- Rutgers/DIMACS Theory of Computing Seminar, February 22nd, 2023.
- CUNY Queens College CS Seminar, February 15th, 2023.
- University of Michigan Theory Seminar, January 27th, 2023.
- Groningen Probability and Statistics Seminar, December 15th, 2022.
- EPFL Optimization Seminar, December 7th, 2022.
- CNRS LIPN lab, December 1st, 2022.
- NYU Theory Seminar, November 17th, 2022.

- *Combinatorial Diameter of Random Polytopes*
  - Discrete Optimization Talks, YouTube, October, 2022.
  - Columbia CS Theory seminar, October 7th, 2022.
  - Cargese Workshop on Combinatorial Optimization, September 23rd, 2022.
  - Workshop *Convex Geometry and its Applications*, MFO Oberwolfach, December 12–18 2021.
  - Symposium on Computational Geometry (SoCG), June 8th, 2022.
  - Online Asymptotic Geometric Analysis Seminar, September 28th, 2021.
  - TU Berlin DISCOGA Research Seminar, September 14th, 2021.
  - Dutch Optimization Seminar, May 27th, 2021.
  - Simons Institute Polymath Project Talks, December 10th, 2020.
- *Why Can We Solve (Integer) Linear Programs?*, STOC TCS Women Spotlight Workshop Rising Star Talk, June 22nd, 2021.
- *On the Integrality Gap of Binary Programs with Gaussian Data*
  - UC Davis Mathematics of Data & Decisions Seminar, February 23rd, 2021.
  - London School of Economics PhD Seminar on Combinatorics, Games and Optimization, February 5th, 2021.
- *Simple Iterative Methods for Linear Optimization over Convex Sets*
  - Integer Programming and Combinatorial Optimization (IPCO), June 27th, 2022.
  - CU Boulder CS Theory Seminar, February 2nd, 2021.
- *Theoretical Analysis of Solving (Integer) Linear Programs*, UC Davis Mathematics of Data & Decisions Seminar, January 19th, 2021.
- *Layered-Least-Squares Interior Point Methods — a Combinatorial Optimization Perspective*
  - CWI N&O Online seminar, May 14th, 2020.
  - TU Munich Applied Geometry and Discrete Mathematics Seminar, March 3rd, 2020.
- *A Friendly Smoothed Analysis of the Simplex Method*
  - Utrecht University Algorithms Seminar, September 3rd, 2019.
  - *Computability in Europe* in Durham, July 15–19, 2019.
  - *SIAM Conference on Applied Algebraic Geometry* in Bern, July 9–13, 2019.
  - *Workshop on Polytopes* in Bochum, March 11–15, 2019.
  - *23rd Combinatorial Optimization Workshop* in Aussois, January 7–11, 2019.
  - Workshop *Convex Geometry and its Applications* at the Mathematisches Forschungsintitut Oberwolfach, December 9–15, 2018.
  - *DIAMANT Symposium* in Veenendaal, November 29–30, 2018.
  - *Ninth Cargese Workshop on Combinatorial Optimization* in Cargese, October 15–19, 2018.
  - *International Symposium on Mathematical Programming* in Bordeaux, July 1–6, 2018.
  - *Symposium on Theory of Computation* in Los Angeles, June 25–29, 2018.
  - *Highlights of Algorithms* in Amsterdam, June 4–6, 2018.